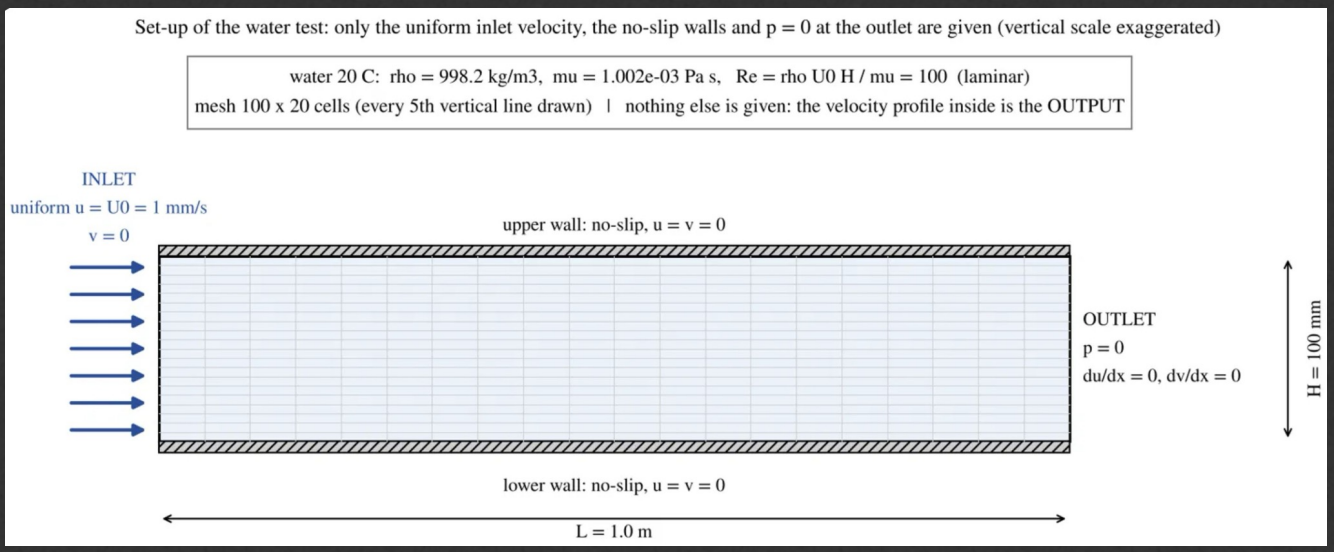
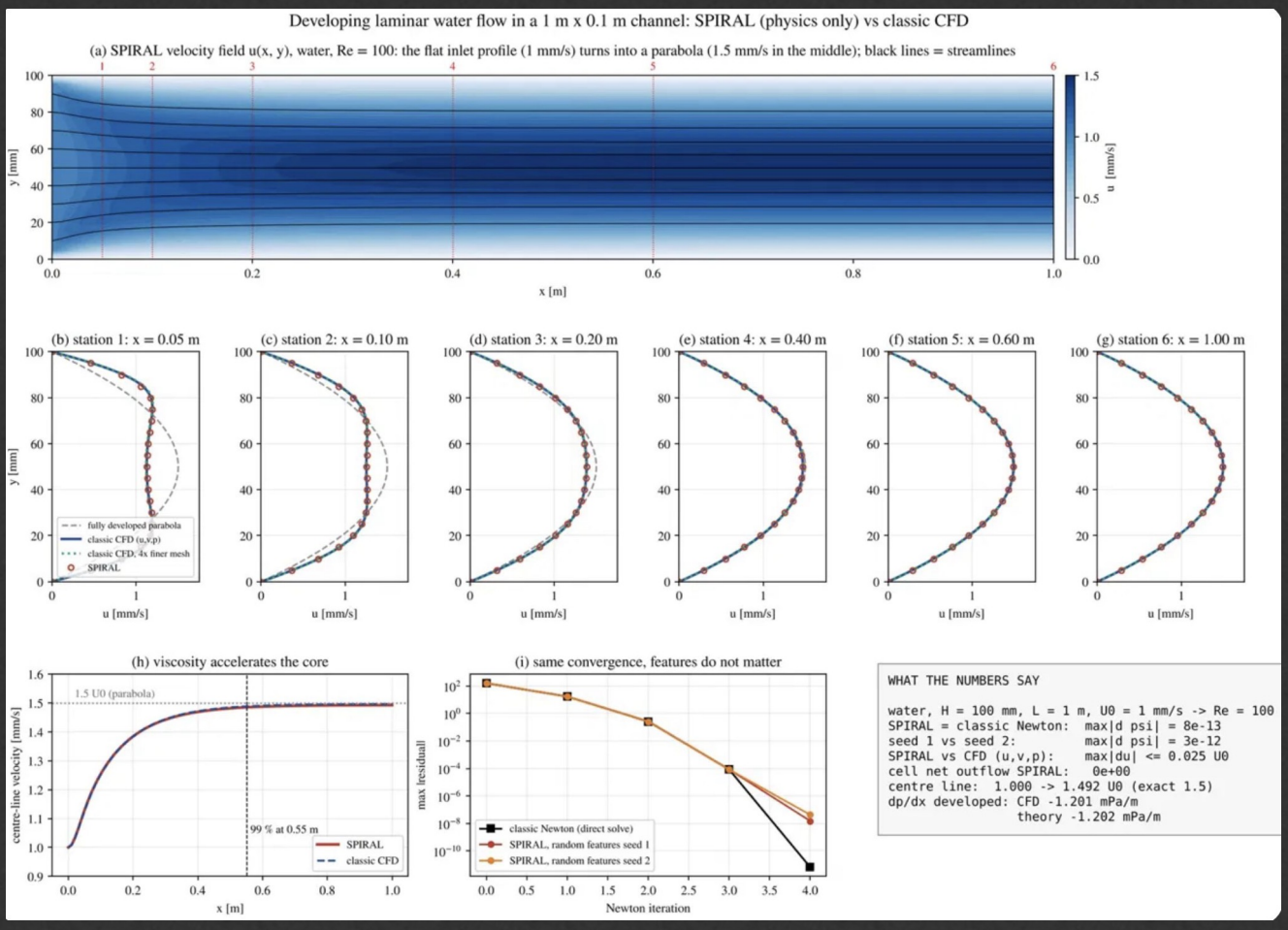
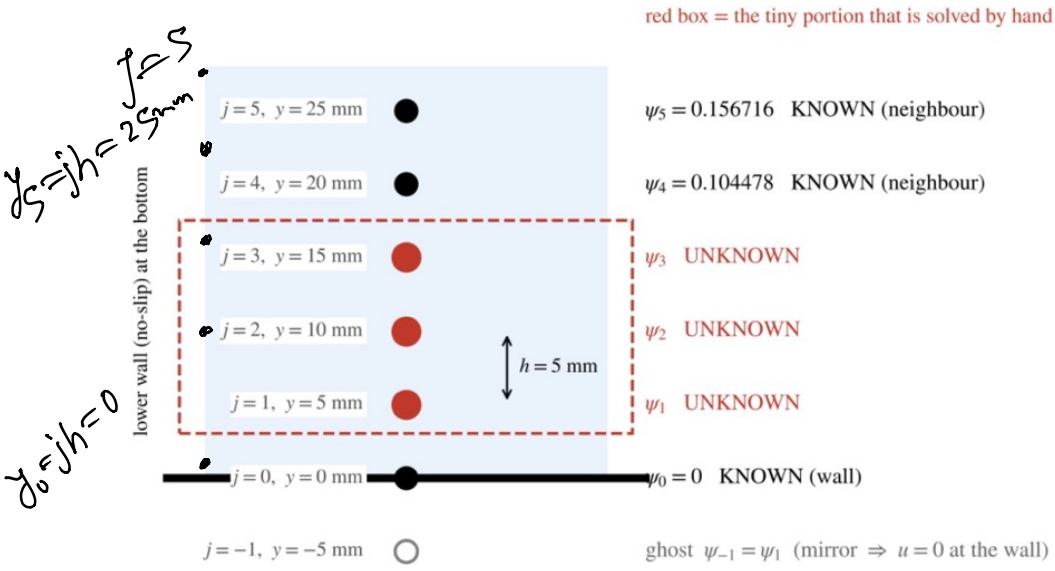
SPIRAL: Stream-function Physics-Informed Random-Feature Learning





How SPIRAL Works: Calibration with 2D Laminar flow

The tiny portion of the real mesh: the 3 nodes next to the wall (mesh of the full run: 20 cells across the gap, $h = H/20 = 5$ mm)



Full Domain 100 X 20 cells.
H

1 cell $h = H/20 = 5$ mm

$\vec{u} = 1$ m/s $\quad Re \simeq 100$ (Laminar)

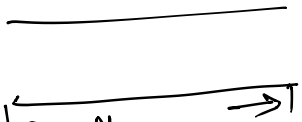
ψ_5, ψ_4 are taken as known from the converged solution to "patch test"

Goal: u at $y = [5, 10, 15]$ mm

No convection here.

Only viscous diffusion of vorticity must balance.

Vorticity is the quantity that carries the shear $\omega = -\partial u/\partial y$



$y \rightarrow$

$0 \le y \le H \quad H = 100$ mm

$u(y) \rightarrow$ velocity along x

$\psi(y) \rightarrow$ stream function $\quad u = \partial\psi/\partial y$, $\psi = 0$ at lower wall

$\psi = \bar{u}H$ (The flow rate at the upper wall)

$\omega(y) \rightarrow$ vorticity (local rotation of the fluid)

$$\omega = -\partial u/\partial y$$

$\mu, \frac{dp}{dx} \rightarrow$ viscosity, pressure gradient

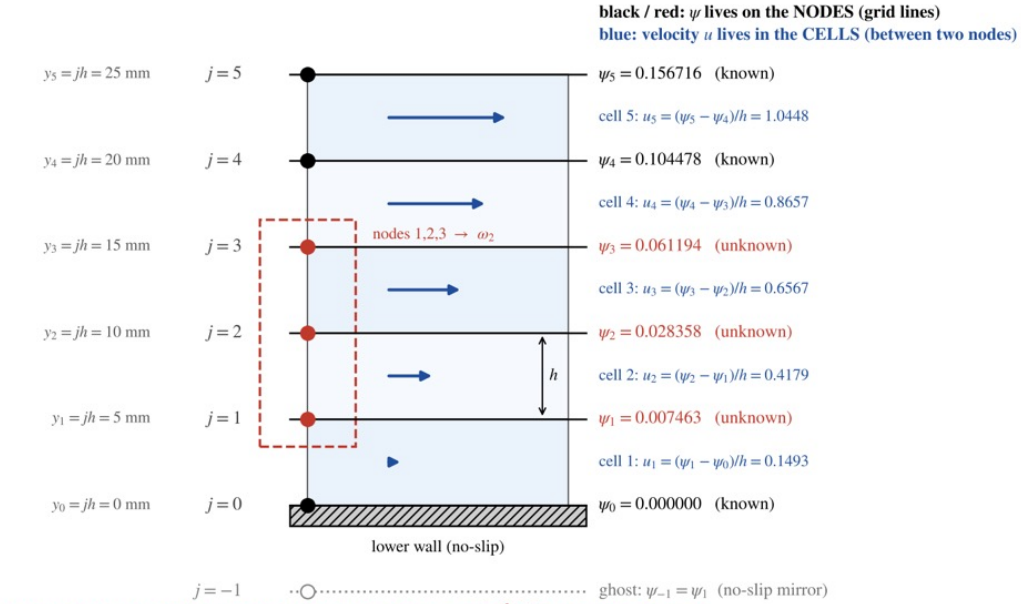
h mesh spacing, $h = H/20 = 5$ mm

$y_j = jh$, $\psi_j, \omega_j \rightarrow$ grid node number j, the values of ψ, ω at that node

$(\psi_j = \psi(y_j))$

$F_j =$ the Residual of the equation at node j ("loss") 0 when the equation satisfied

Notation: ψ_2 (number = NODE) is the VALUE of ψ at $y = 2h = 10$ mm. $\psi_y = \partial\psi/\partial y$ (letter y = DERIVATIVE).



dashed box = the three nodes used for the vorticity at node 2: $\omega_2 = -\frac{\psi_1 - 2\psi_2 + \psi_3}{h^2} = -400\,(\psi_1 - 2\psi_2 + \psi_3)$

numbers: $-400\,(0.007463 - 2 \cdot 0.028358 + 0.061194) = -400 \cdot 0.011941 = -4.7764$ (units $\bar{u}/H$)

$u_y = \partial u/\partial y$ (first derivative) $u_{yy} = \partial^2 u/\partial y^2$ (second derivative)

1. "fully Developed" gives

far downstream nothing changes along x and $v=0$

The x-momentum equation has no convection and reduces to a balance between pressure and viscous force.

$$0 = -\frac{dp}{dx} + \mu \frac{\partial^2 u}{\partial y^2} \Rightarrow u_{yy} = \frac{1}{\mu}\frac{dp}{dx} = \text{constant} \quad \text{---(1)}$$

where: $u = \psi_y$

$$\omega = -u_y = -\psi_{yy}$$

(1) differentiable wrt y

$$u_{yyy} = 0 \Rightarrow \omega_{yy} = -u_{yyy} \Rightarrow \psi_{yyyy} = 0 \quad \text{---(2)}$$

vorticity must vary linearly with y

2. Putting it on mesh

second derivative of any function f at node j from its neighbours (Taylor series error $O(h^2)$)

$$f_{yy}(y_j) \simeq \frac{f_{j-1} - 2f_j + f_{j+1}}{h^2} \quad \text{---(3)}$$

$h = 5mm = 0.05m$

A. Vorticity from ψ ($1/h^2 = 400$): $\omega_j = -\psi_{yy}(y_j)$

$$\simeq -\frac{\psi_{j-1} - 2\psi_j + \psi_{j+1}}{h^2} = -400(\psi_{j-1} - 2\psi_j + \psi_{j+1})$$

B. Equation $\omega_{yy} = 0$: $\omega_{j-1} - 2\omega_j + \omega_{j+1} = 0$
at unknown nodes $j = 1, 2, 3$

3. Stage A: The five vorticities

known, $\psi_0 = 0$ (wall)

ψ_4, ψ_5 (neighbours from the converged solution)

$\omega_j = -400\, b_j \quad (b_j = \omega_{j-1} - 2\psi_j + \psi_{j+1})$

$b_1 = \psi_0 - 2\psi_1 + \psi_2 = -2\psi_1 + \psi_2$

$b_2 = \psi_1 - 2\psi_2 + \psi_3$

$b_3 = \psi_2 - 2\psi_3 + \psi_4$

$b_4 = \psi_3 - 2\psi_4 + \psi_5$

The wall node $j = 0$. The formula needs ψ_{-1}, a point outside the fluid (a "ghost").

no slip wall $u = \psi_y = 0$

central difference $(\psi_1 - \psi_{-1})/2h = 0$

hence, $\psi_{-1} = \psi_1$ and

$b_0 = \psi_{-1} - 2\psi_0 + \psi_1 = \psi_1 - 0 + \psi_1 = 2\psi_1$

$\omega_0 = -800\psi_1$

Stage B: Build $w_{j-1} - 2w_j + w_{j+1}$ node by node

Every w contains the factor -400

The equation at node j reads $-400(b_{j-1} - 2b_j + b_{j+1}) = 0$

Node 1: $E_1 = b_0 - 2b_1 + b_2$

Node 1: $E_1 = b_0 - 2b_1 + b_2$					
term	ψ_1	ψ_2	ψ_3	ψ_4	ψ_5
$b_0 = 2\psi_1$	2				
$-2b_1 = -2(-2\psi_1 + \psi_2)$	4	-2			
$b_2 = \psi_1 - 2\psi_2 + \psi_3$	1	-2	1		
sum E_1	**7**	**−4**	**1**		

Node 2: $E_2 = b_1 - 2b_2 + b_3$					
b_1	-2	1			
$-2b_2$	-2	4	-2		
b_3		1	-2	1	
sum E_2	**−4**	**6**	**−4**	**1**	

Node 3: $E_3 = b_2 - 2b_3 + b_4$					
b_2	1	-2	1		
$-2b_3$		-2	4	-2	
b_4			1	-2	1
sum E_3	**1**	**−4**	**6**	**−4**	**1**

Results: $E_1 = 7\psi_1 - 4\psi_2 + \psi_3$

$E_2 = -4\psi_1 + 6\psi_2 - 4\psi_3 + \psi_4$

$E_3 = \psi_1 - 4\psi_2 + 6\psi_3 - 4\psi_4 + \psi_5$

The pattern 1, −4, 6, −4, 1 is the discrete form of ψ_{yyyy} (second difference of a second difference) It reaches two nodes to each side which is why ψ_4, ψ_5 enter. The first row has 7 instead of 6 only because of the wall ghost ($\psi_{-1} = \psi_1$).

5 Check it with numbers

ψ used	E_1	E_2	E_3
plug-flow guess $\psi_{1,2,3} = 0.05, 0.10, 0.15$ (with ψ_4, ψ_5 as given)	0.1000	−0.0955	0.2888
converged network result	≈ 0	≈ 0	≈ 0
$\psi_{1..5} = 0.007463, 0.028358, 0.061194, 0.104478, 0.156716$			
exact parabola $\psi = 3\eta^2 - 2\eta^3$, $\eta = y/H$:	−0.0005	0	0
$\psi_{1..5} = 0.00725, 0.028, 0.06075, 0.104, 0.15625$			

$7(0.05) - 4(0.10) + 0.15 = 0.1$

It is not 0 but the network job is to drive E to zero.

The exact parabola gives $E_2 = E_3 = 0$
(a cubic has $\psi_{yyyy} = 0$ and five-point formula is exact for cubics)
but, $E_1 = 7(0.00725) - 4(0.028) + 0.06075 = -0.0005$
due to the wall row with it's mirror ghost closure.
can not produce the exact solution.
That is 0.5% mismatch between SPIRAL network the exact parabola.

So, $E = S\psi + c$ $\quad S = \begin{bmatrix} 7 & -4 & 1 \\ -4 & 6 & -4 \\ 1 & -4 & 6 \end{bmatrix}$

$c = (0, \psi_4, -4\psi_4 + \psi_5)^T$

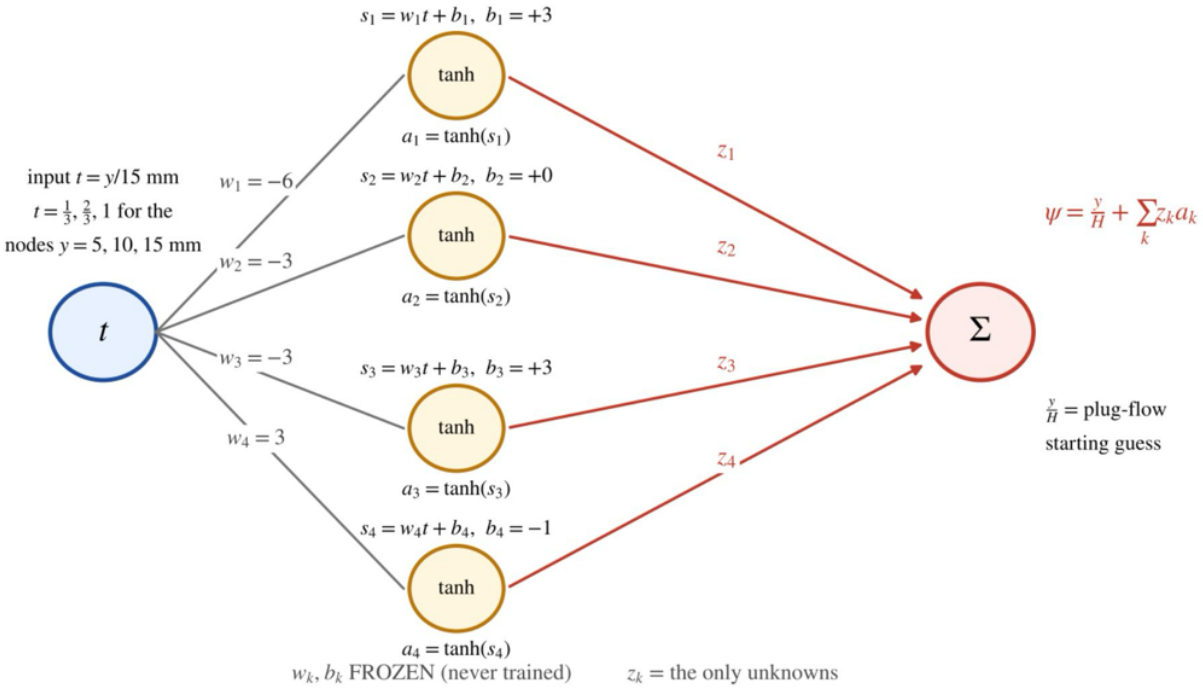
$L = \frac{1}{2}\sum_j F_j^2$ $\quad$ starting guess = plug flow

$\psi^T = y/H \quad (0.05, 0.10, 0.15)$

(all vorticities are 1)

$E = \begin{bmatrix} 0.1000000 \\ -0.095522 \\ 0.288806 \end{bmatrix}$ $\quad L = 0.005267$

Network of the unit test: 1 input, 4 hidden tanh nodes (3 equations ⇒ 4 hidden nodes), 1 output



Step 2 : Forward Pass

Network input $t = y/15$ mm

hidden nodes $s_k = w_k t + b_k$ $\quad a_k = \tanh s_k$

output $\psi(t) = y/H + \sum_k z_k a_k$

weights $w = (-6, -3, -3, 3)$
$b = (3, 0, 3, -1)$
} Easy for hand calculation

with $z = 0$ the output is the plug flow

	s_1	s_2	s_3	s_4
$t = \frac{1}{3}$ (5 mm)	1	−1	2	0
$t = \frac{2}{3}$ (10 mm)	−1	−2	1	1
$t = 1$ (15 mm)	−3	−3	0	2

$$\Phi = [\tanh s_k] = \begin{bmatrix} 0.7616 & -0.7616 & 0.9640 & 0.0000 \\ -0.7616 & -0.9640 & 0.7616 & 0.7616 \\ -0.9951 & -0.9951 & 0.0000 & 0.9640 \end{bmatrix}$$

Step 3: The SPIRAL Training step
Network in kernel form

Chain rule residual→ $\psi \to z$: $J = S\Phi$ (3 equations, 4 weights z; there are more weights than equations, so a minimum-norm solution exists): $J\Delta z = -E$ is solved by $\Delta z = J^\top \alpha$ with $(JJ^\top)\alpha = -E$.

$$J = S\Phi = \begin{bmatrix} 7.382 & -2.470 & 3.702 & -2.082 \\ -3.636 & 1.242 & 0.713 & 0.713 \\ -2.162 & -2.876 & -2.082 & 2.738 \end{bmatrix}, \qquad K = JJ^\top = \begin{bmatrix} 78.64 & -28.75 & -22.27 \\ -28.75 & 15.78 & 4.76 \\ -22.27 & 4.76 & 24.78 \end{bmatrix}$$

Gauss Elimination 3x3 system $K\alpha = -E$

$$\alpha = \begin{bmatrix} -0.006059 \\ 0.000178 \\ -0.017136 \end{bmatrix} \quad (\text{check: } K\alpha = \begin{bmatrix} -0.100000 \\ 0.095522 \\ -0.288806 \end{bmatrix} = -E \ \checkmark)$$

$$\Delta z = J^\top \alpha = \begin{bmatrix} -0.00832 \\ 0.06447 \\ 0.01338 \\ -0.03417 \end{bmatrix}, \qquad \psi = \frac{y}{H} + \Phi\,\Delta z = \begin{bmatrix} 0.007463 \\ 0.028358 \\ 0.061194 \end{bmatrix}, \qquad E = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$

Step 9: Output Comparison

Velocity at the nodes (central difference of the output): $u_j = (\psi_{j+1} - \psi_{j-1})/(2h) = 10\,(\psi_{j+1} - \psi_{j-1})$:

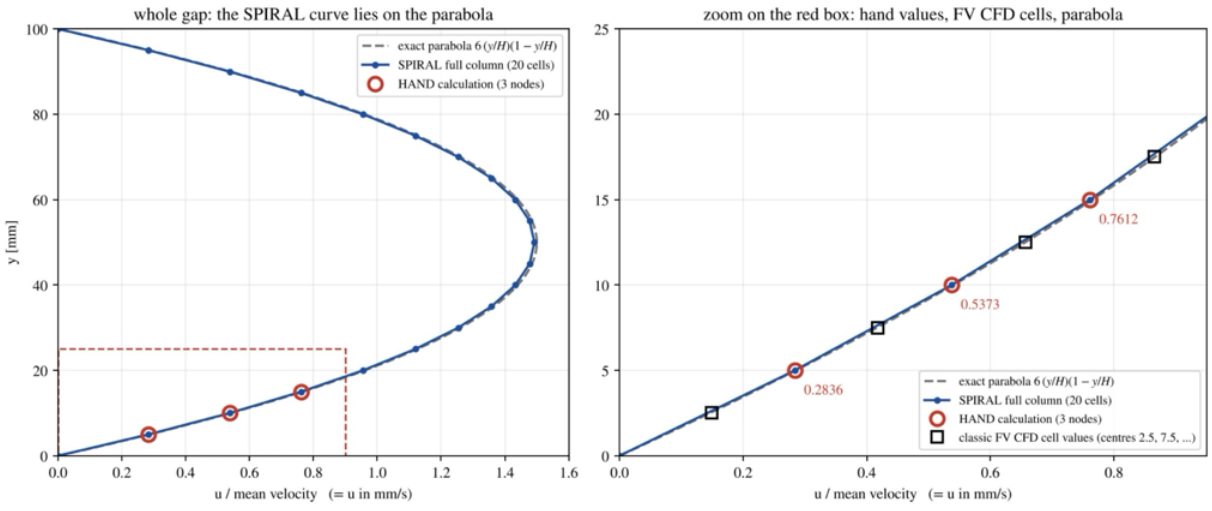
$$u(5\text{ mm}) = 10\,(0.02836 - 0) = 0.28358$$
$$u(10\text{ mm}) = 10\,(0.06119 - 0.00746) = 0.53731$$
$$u(15\text{ mm}) = 10\,(0.104478 - 0.02836) = 0.76119$$

Three independent references: (a) the *full* SPIRAL solution on all 20 cells of the column, (b) the *2-D water run* (100×20 mesh, SPIRAL) read at $x = 1$ m, (c) the exact parabola.

y [mm]	hand (this sheet)	full 1-D column	full 2-D run, $x = 1$ m	exact $6\frac{y}{H}(1 - \frac{y}{H})$
5	0.28358	0.28358	0.28368	0.28500
10	0.53731	0.53731	0.53747	0.54000
15	0.76119	0.76119	0.76137	0.76500

Classic finite-volume CFD on the same 20 cells, no network: momentum balance per cell, $G = \frac{dp/dx}{\mu}$: $-3u_1 + u_2 = Gh^2$ (wall cell), $u_{k-1} - 2u_k + u_{k+1} = Gh^2$, $\sum_k u_k h = 1$. Its cell velocities equal the network's $(\psi_k - \psi_{k-1})/h$:

cell (centre)	1 (2.5 mm)	2 (7.5 mm)	3 (12.5 mm)	4 (17.5 mm)
SPIRAL $= (\psi_k - \psi_{k-1})/h$	0.149254	0.417910	0.656716	0.865672
classic FV CFD	0.149254	0.417910	0.656716	0.865672



Different hidden weights but same answer

Network B: $w = (-6, -3, -3, 6)$, $b = (5, 4, 5, -2)$. Then $\Delta z_B = \begin{bmatrix} 0.0147 \\ -0.0260 \\ -0.0314 \\ -0.0276 \end{bmatrix}$ while $\Delta z_A = \begin{bmatrix} -0.0083 \\ 0.0645 \\ 0.0134 \\ -0.0342 \end{bmatrix}$,

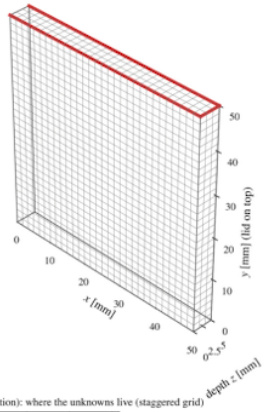
but $\psi_B = \begin{bmatrix} 0.00746 \\ 0.02836 \\ 0.06119 \end{bmatrix} = \psi_A$. The physics fixes ψ; the network is only a parametrisation.

Advection-Diffusion of a passive-Tracer in a 3D Lid-Driven cavity $Re = 1000$

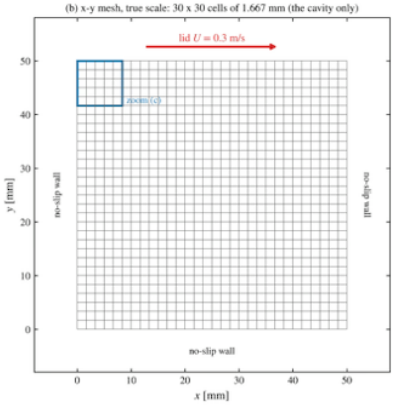
SPIRAL-3D mesh, true dimensions: 30 x 30 x 3 = 2,700 hexahedral cells, 3,844 nodes, cell 1.667 x 1.667 x 1.667 mm

flow unknowns: 6,003 vector-potential edge values (+ 2,700 cell values of phi, which starts at 0 and is released later as a drop) | ANSYS Reference mesh: 100 x 100 x 10 = 100,000 cells (0.5 mm), 112,211 nodes (37x more cells)

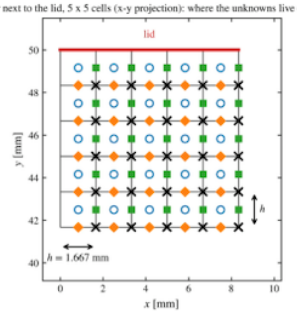
(a) the slab 50 x 50 x 5 mm (true proportions), mesh on the front, right and top faces; red edges = moving lid (top wall)

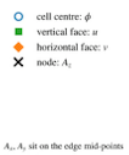


(b) x-y mesh, true scale: 30 x 30 cells of 1.667 mm (the cavity only)



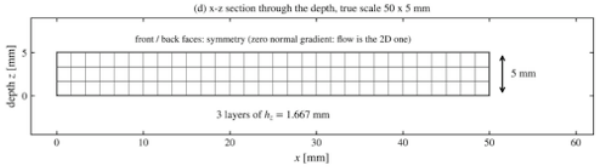
(c) corner next to the lid, 5 x 5 cells (x-y projection): where the unknowns live (staggered grid)





A_z, A_x sit on the edge mid-points through the u, v points; u, v = curl A (divergence-free to round-off)

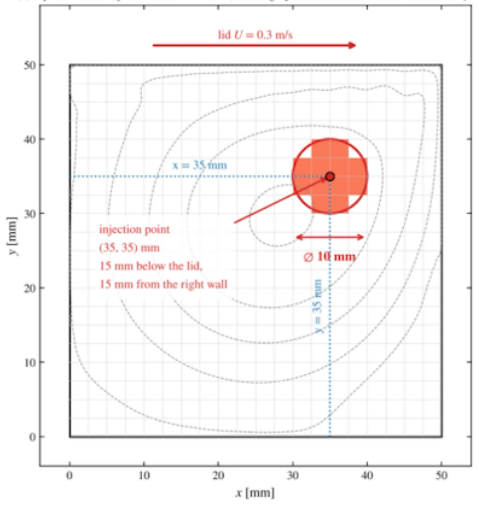
(d) x-z section through the depth, true scale 50 x 5 mm



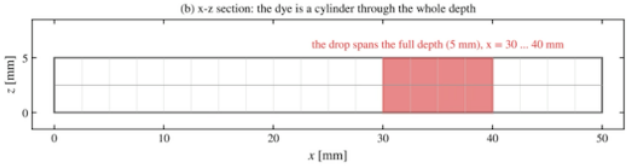
Dim 50X50X5 mm grid size = 30X30X3

SPIRAL-3D: injection of the dye drop into the cavity flow, Re = 1000, 20×20×2 cells. The circle is not part of the domain: it is only the initial condition of ϕ.

(a) x-y view at the injection time t = 6.10 s (dashed grey: streamlines of the flow, red cells: $\phi = 1$)



(b) x-z section: the dye is a cylinder through the whole depth



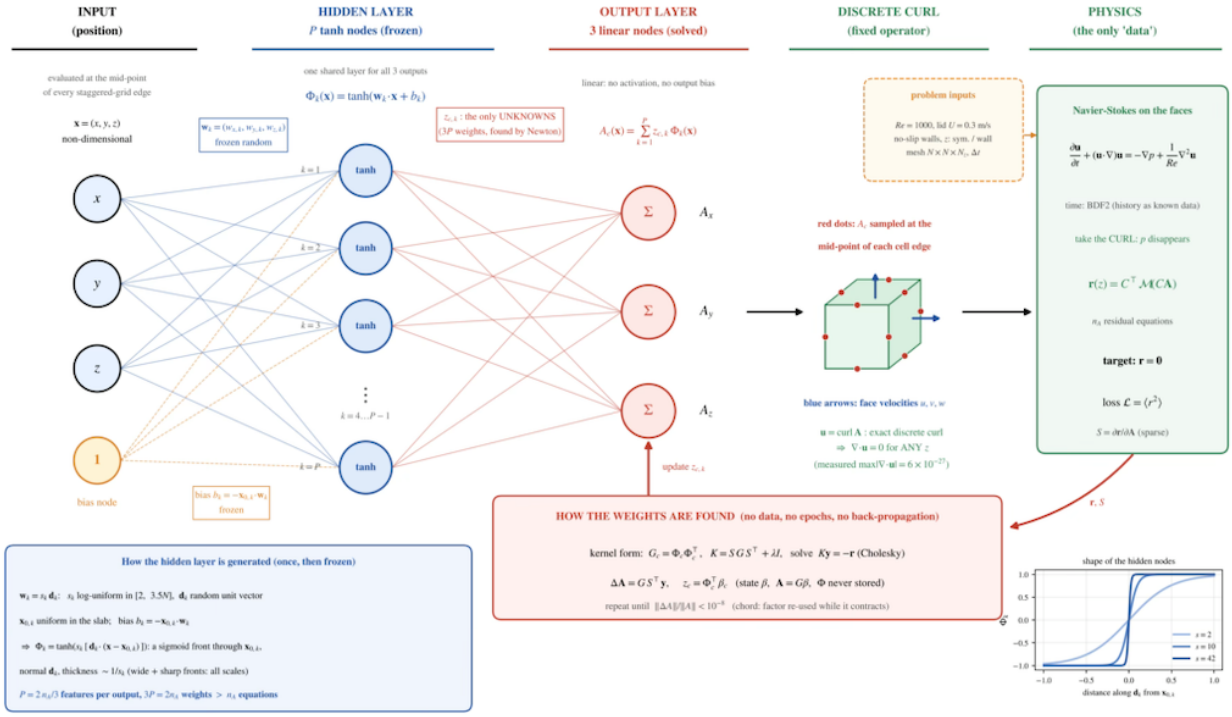
(c) injection set-up

injection time	t = 6.10 s (the moment the kinetic energy is flat)
injection point (centre)	x = 35 mm, y = 35 mm, all z (0.70 L, 0.70 L)
circle diameter	10 mm (radius 5 mm = 2.0 cells)
extent through the depth	full depth, 5 mm (cylinder volume 392.7 mm^3)
cells with $\phi = 1$	12 per layer, 24 in 3D (volume 375.0 mm^3 on this mesh)
initial field	$\phi = 1$ inside the circle, $\phi = 0$ everywhere else
diffusivity / Peclet	D = 2.50e-05 m^2/s, Pe = 600
followed for	4.0 s (to t = 10.10 s)

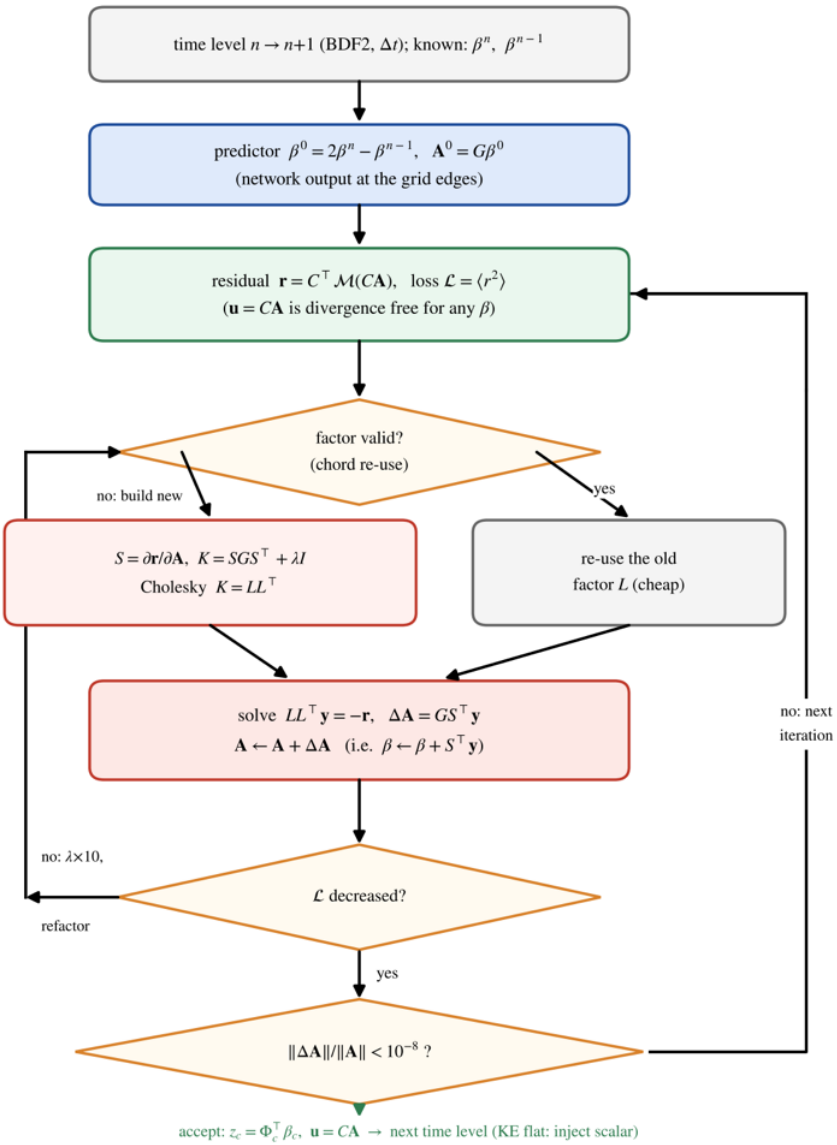
Inject $\phi = 1$ when Kinetic Energy is stable

We have used the Same Configurations of rCFD

The Network architecture

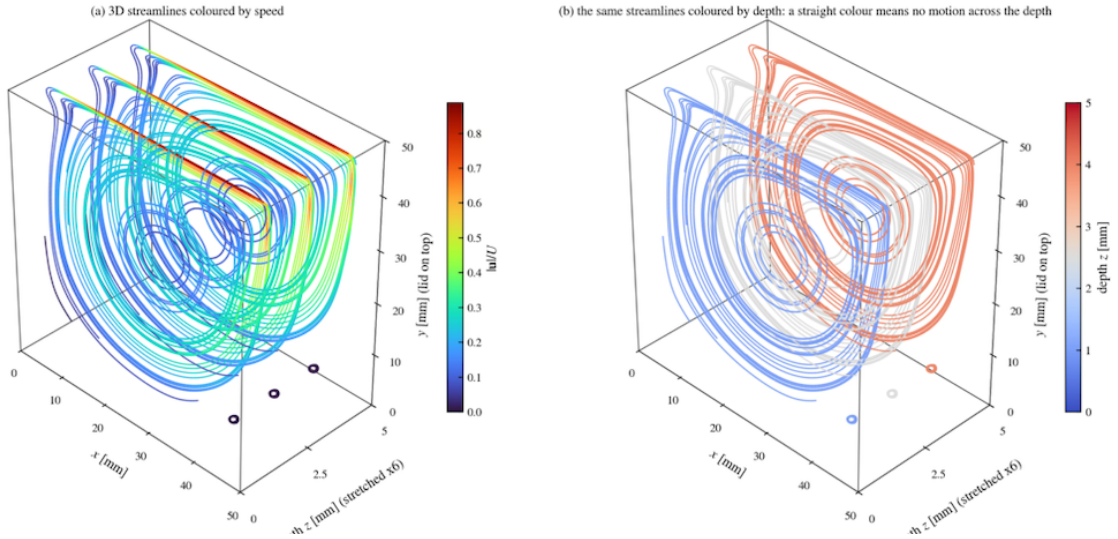


One time step of SPIRAL: Newton / Levenberg-Marquardt in kernel form



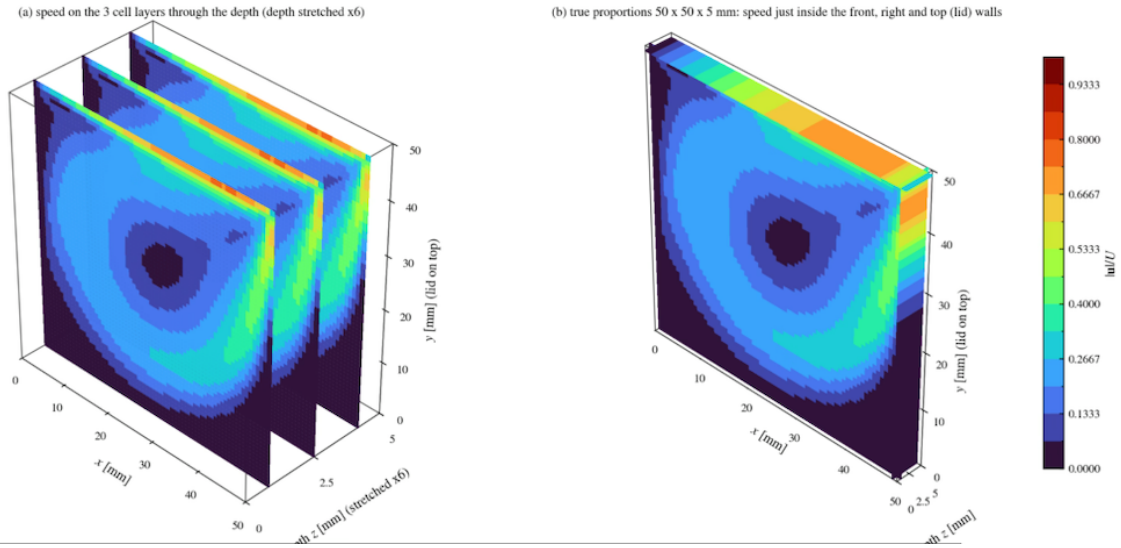
Results and validation

SPIRAL-3D streamlines, isometric view (depth stretched x6), t = 8.14 s, Re = 1000, front/back faces: symmetry, max $|w|/U$ = 9.9e-11



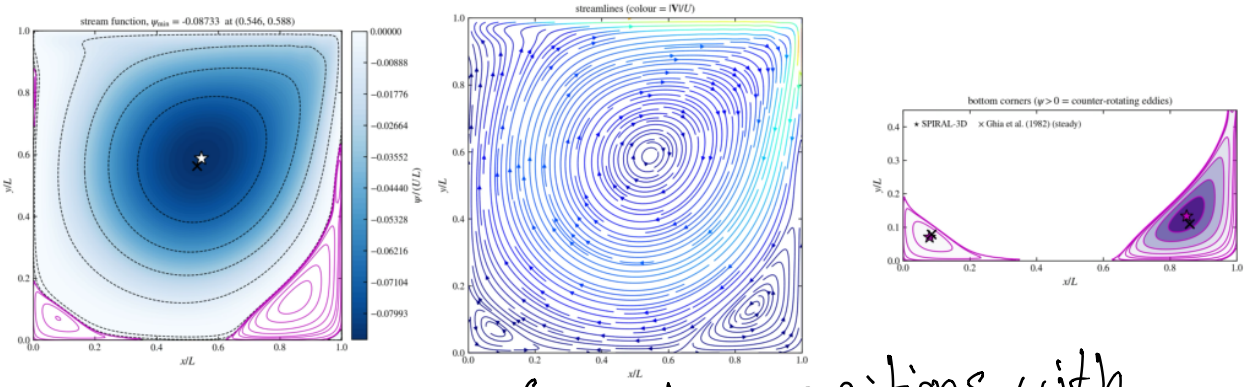
3D streamlines

SPIRAL-3D velocity magnitude, isometric view, t = 8.14 s, Re = 1000, 30×30×3 cells, front/back faces: symmetry [max $|w|/U$ = 9.9e-11, depth variation of u = 1.7e-10 U]

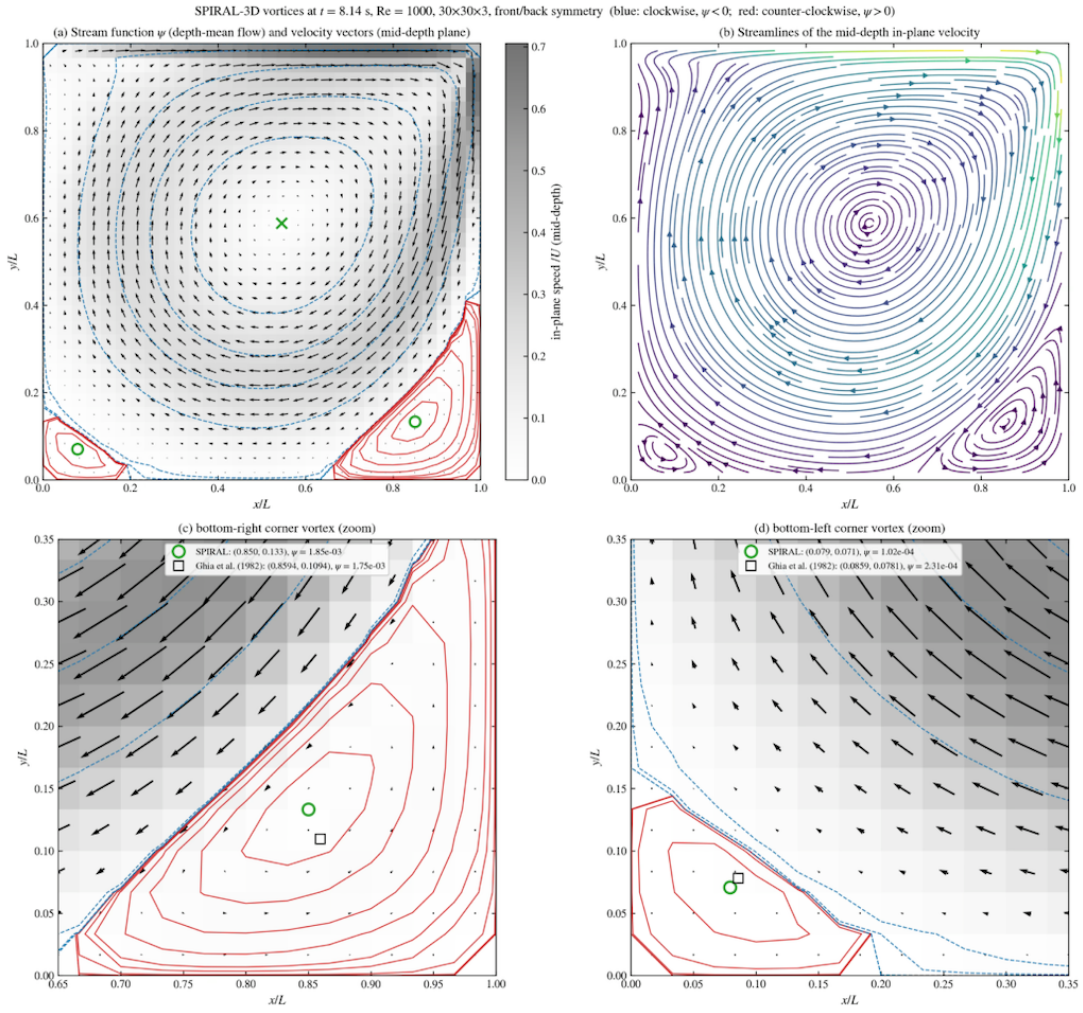


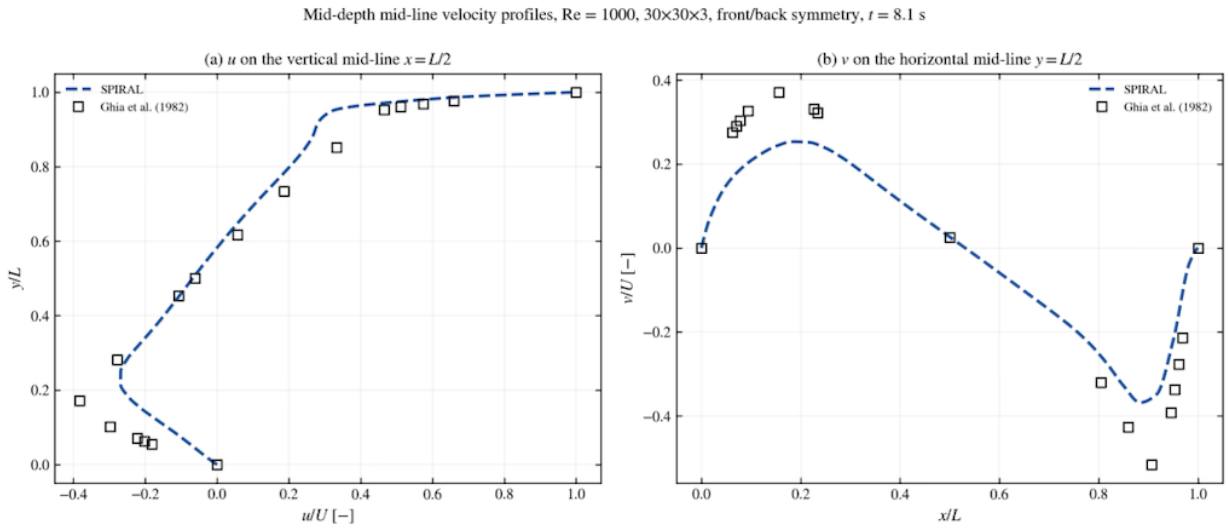
Velocity Contours

Cavity-flow check: 30×30×3 cells, front/back symmetry, Re = 1000, t = 8.1 s. Black contours: primary vortex, magenta: corner eddies (Ghia levels)



validation of vorties positions with Ghia et. al 1982

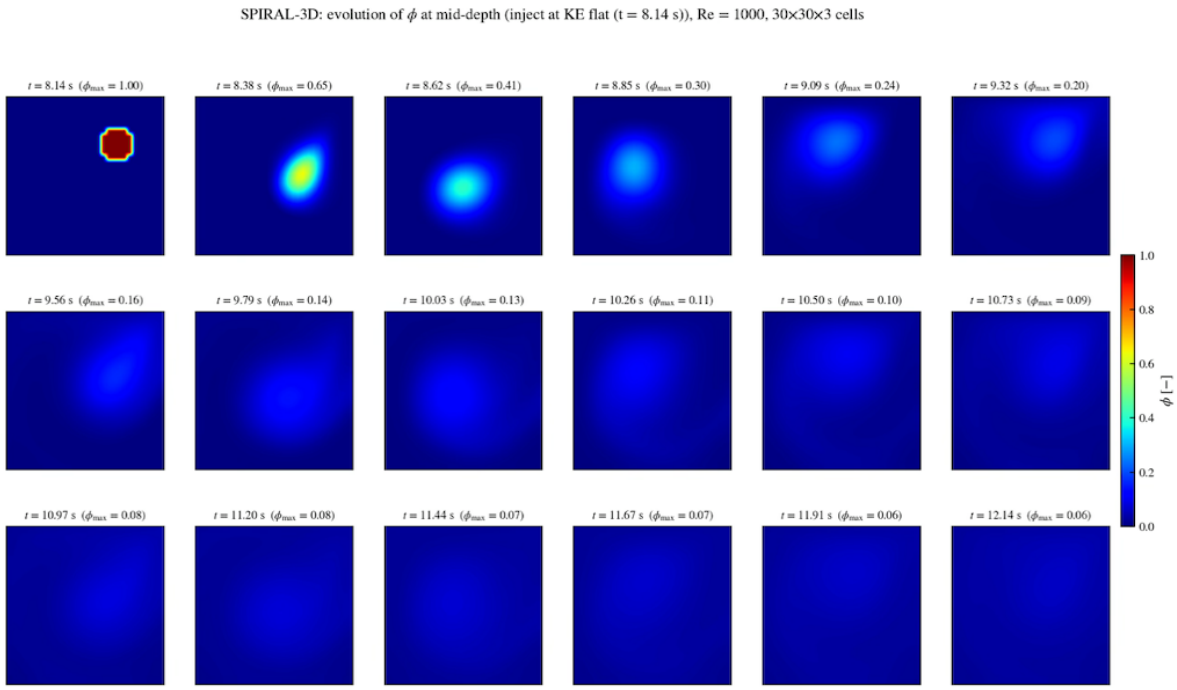




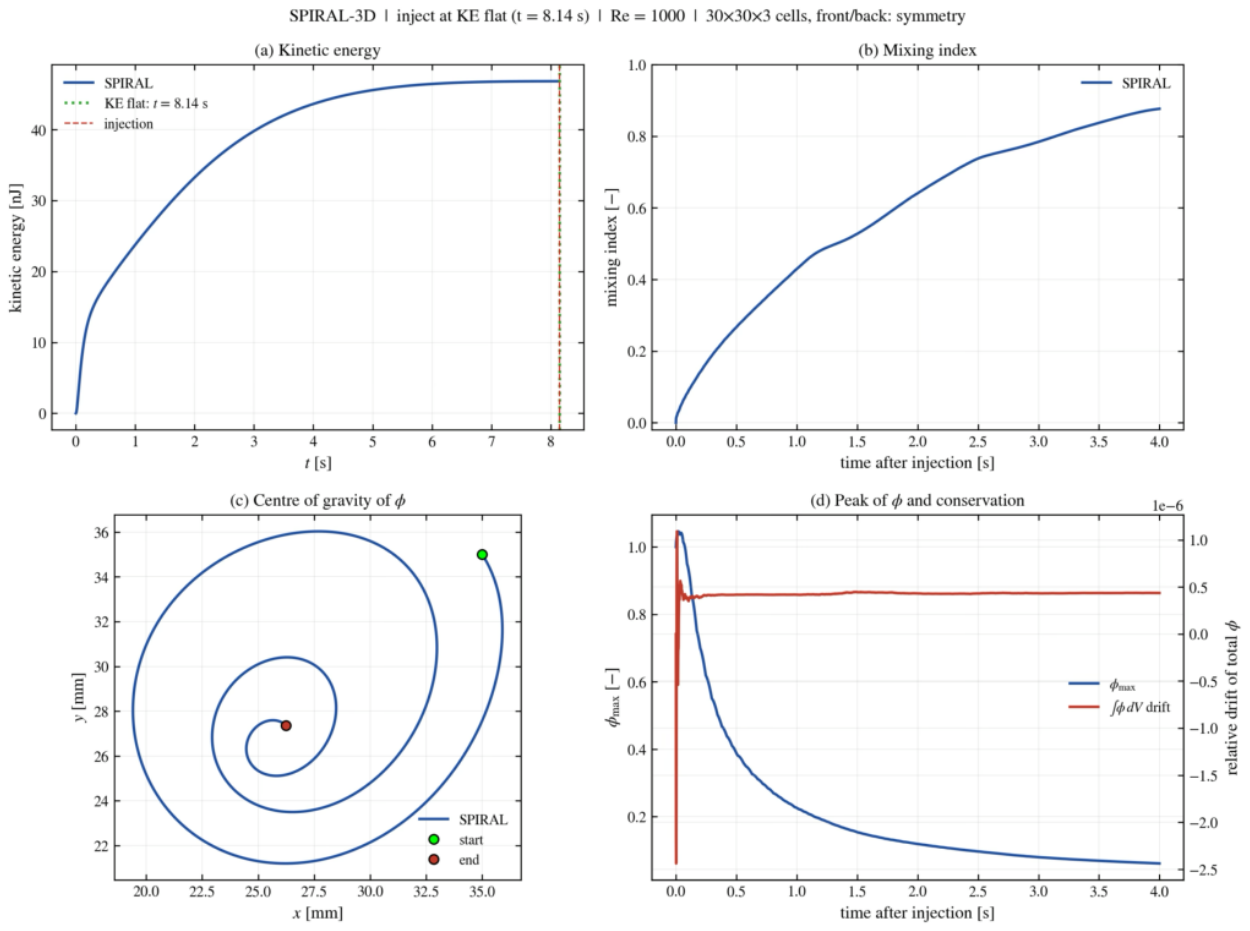
Uavg

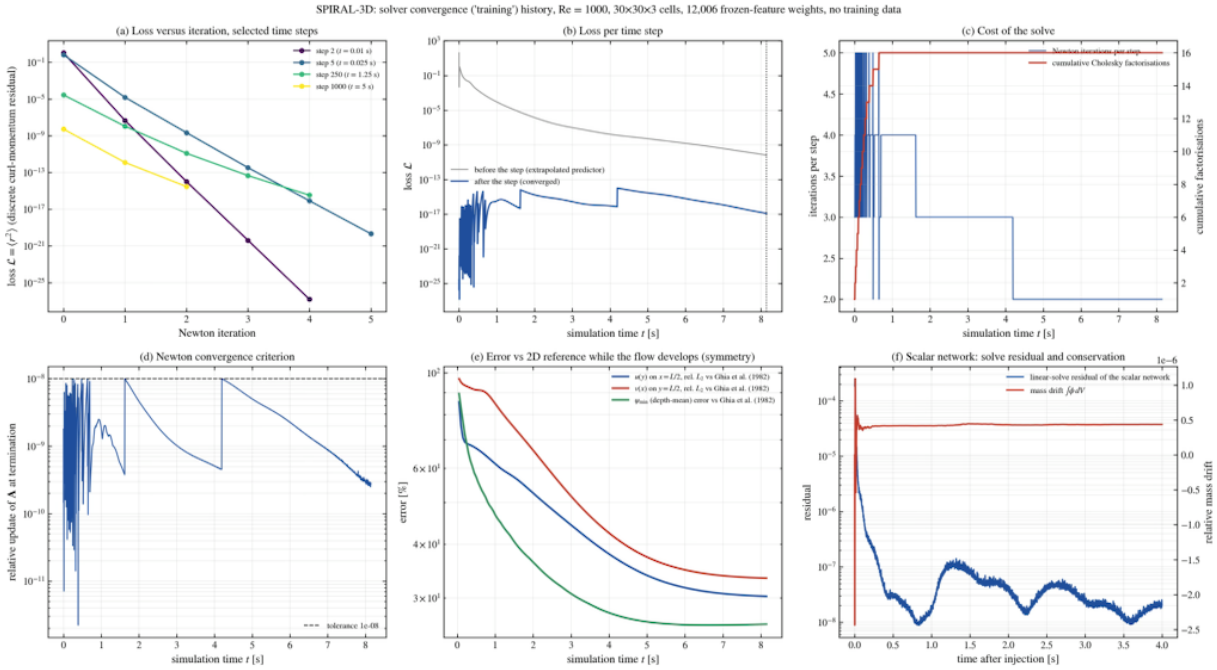
Wavg

✓ acceptable agreement with reference paper

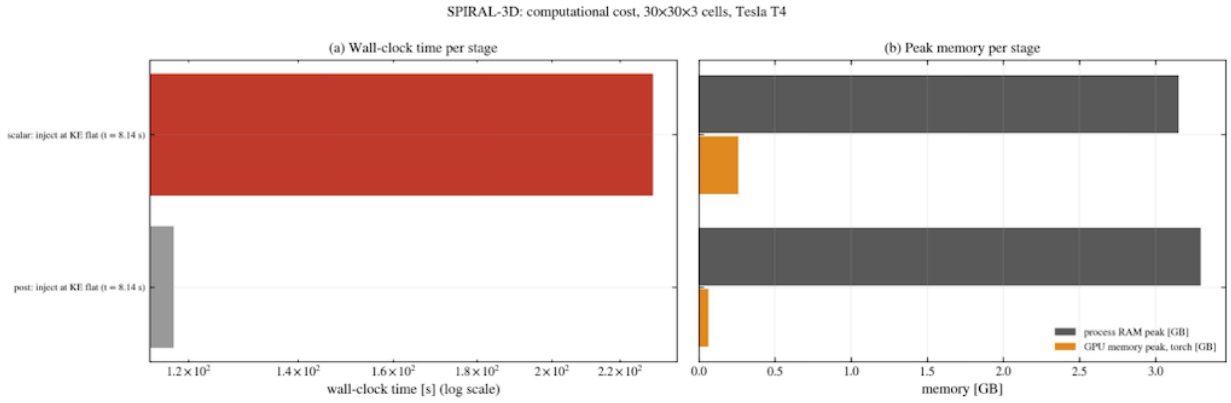


Real-Time mixing sequences





Training History



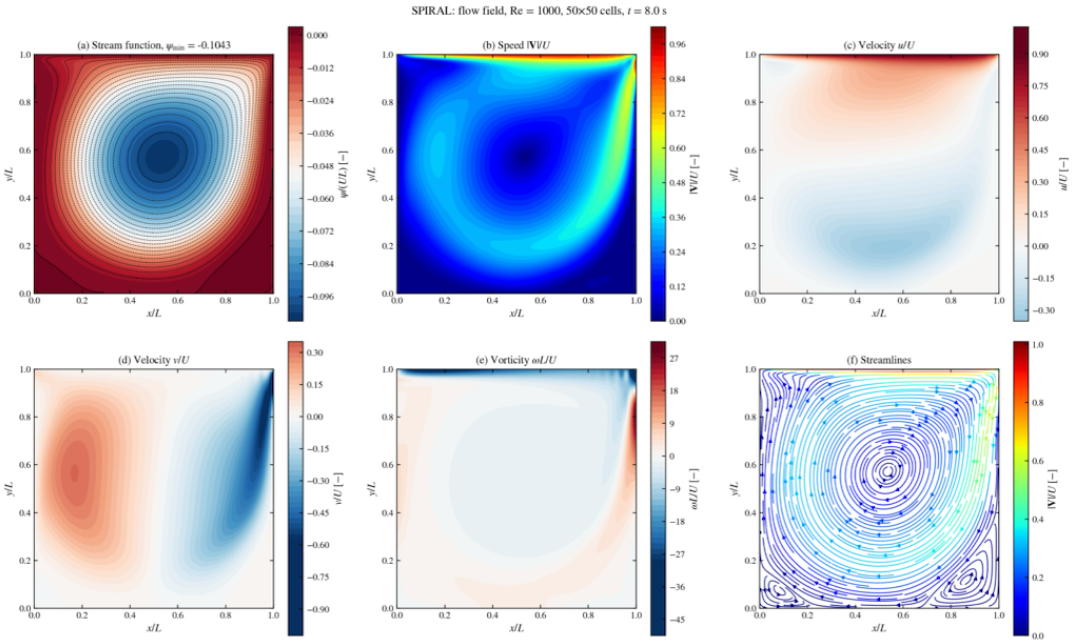
Solution Time Resources

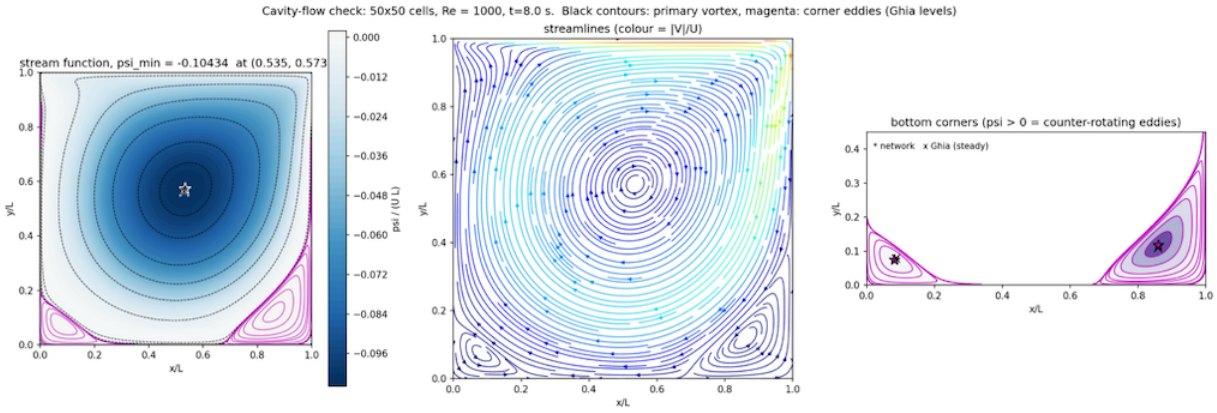
Comparison with rCFD:

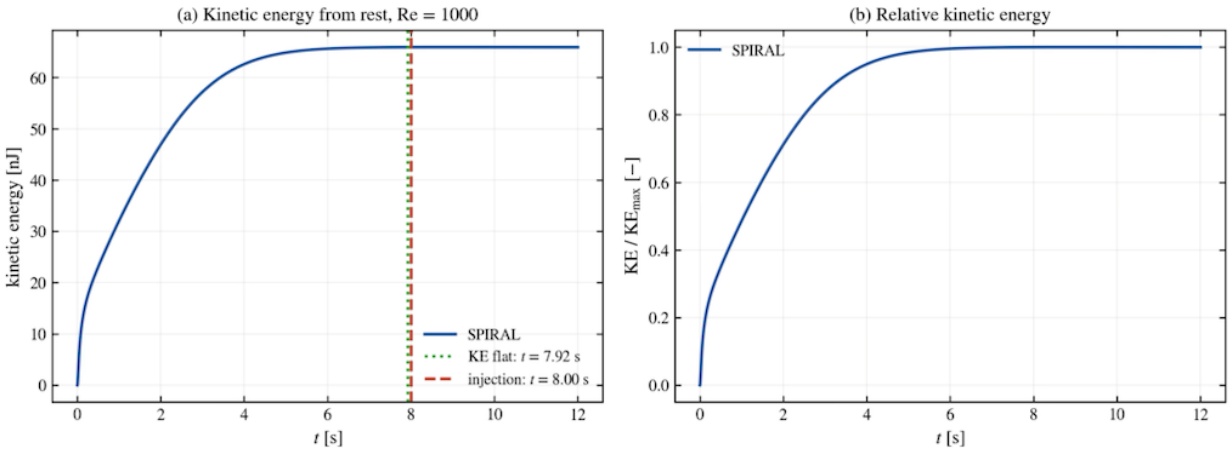
	SPIRAL	rCFD
Database	No	Yes
Stages	1 (Run code)	3: (spin-up), rCFD Database, rCFD predict
Total Time:	347 sec	≈ 6 hours
Prediction Time:	117s	<10 sec

verdict: SPIRAL wins ✓

2D Lid-Driven Cavity Result Calibration with ANSYS Results

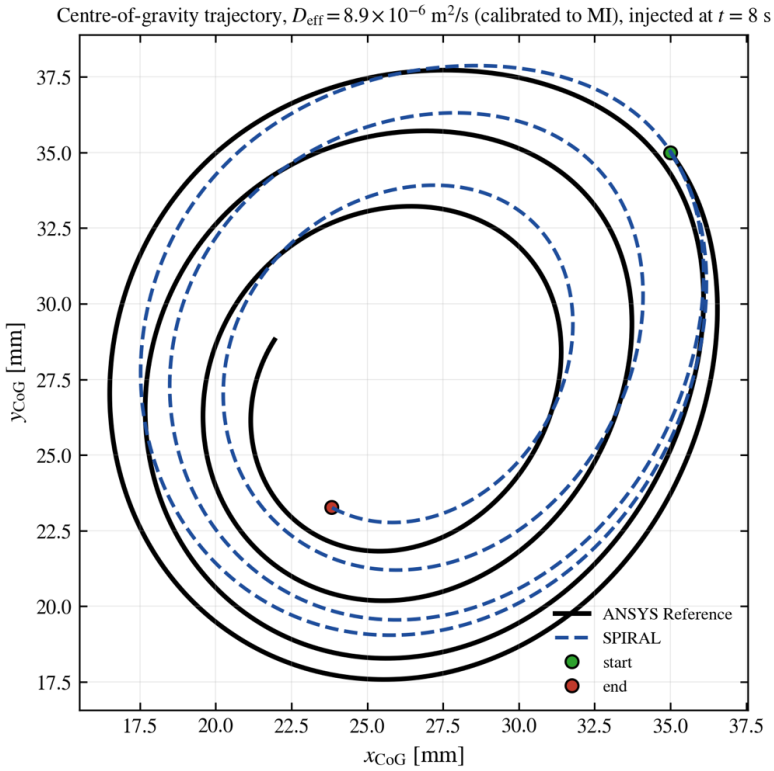


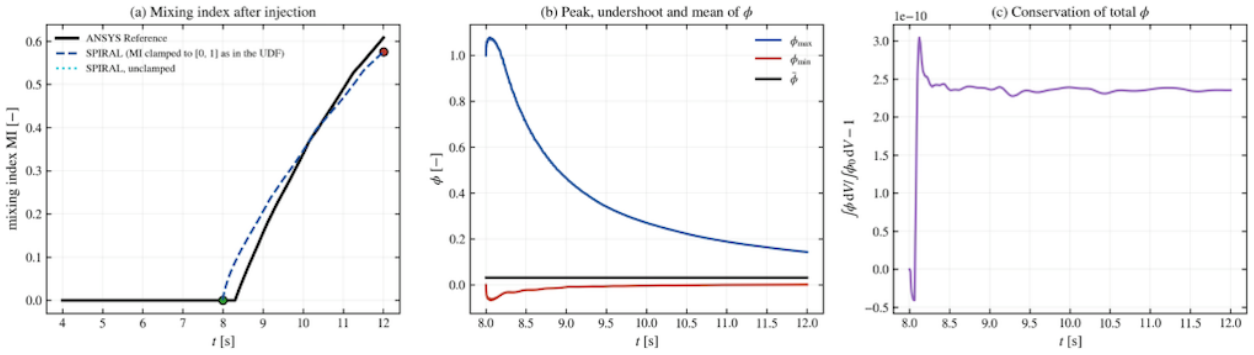


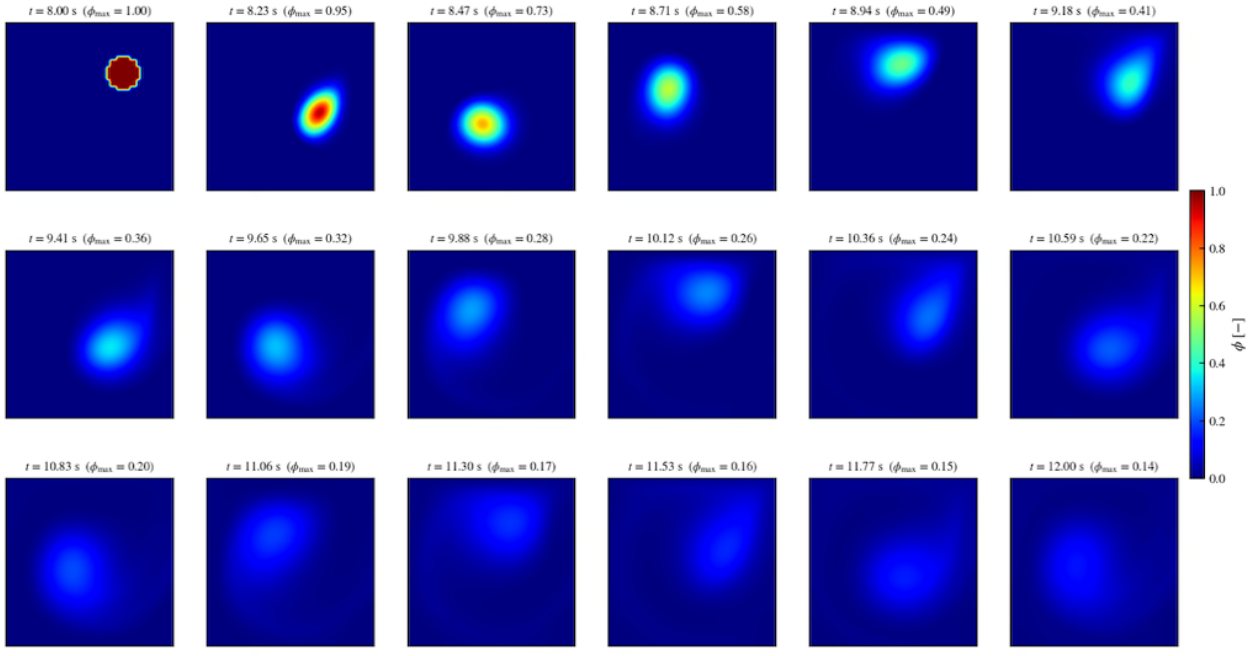


Calibrated

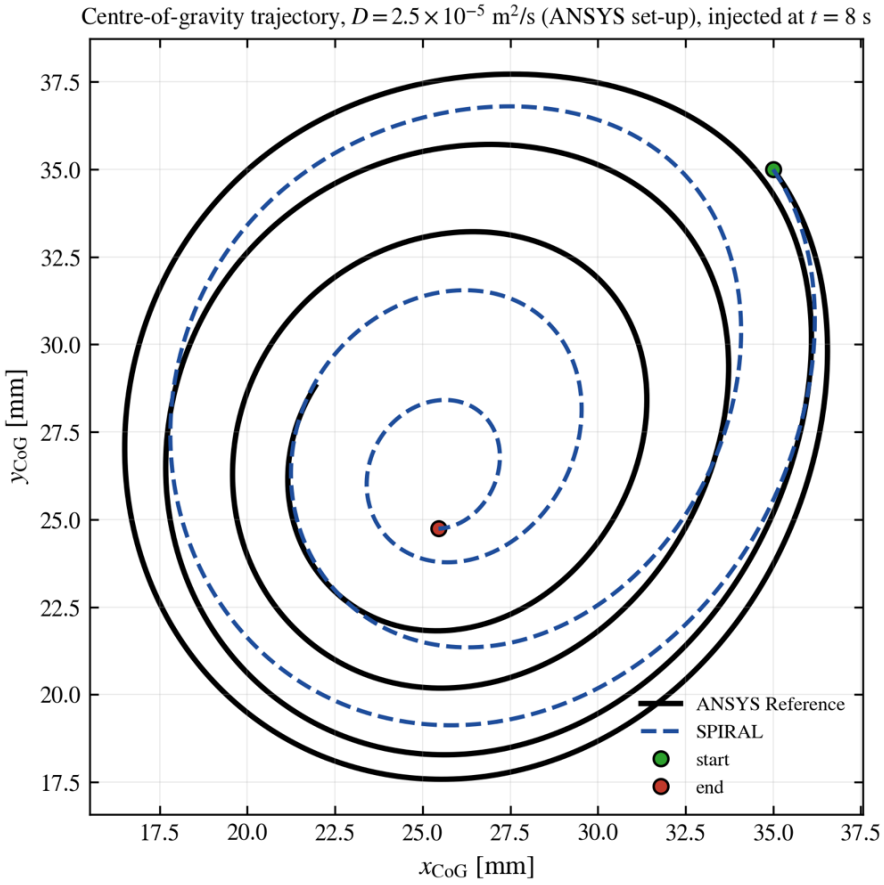
The Network automatically fitted the diffusivity $= 8.9 \times 10^{-6}\ m^2/s$ to closely match ANSYS

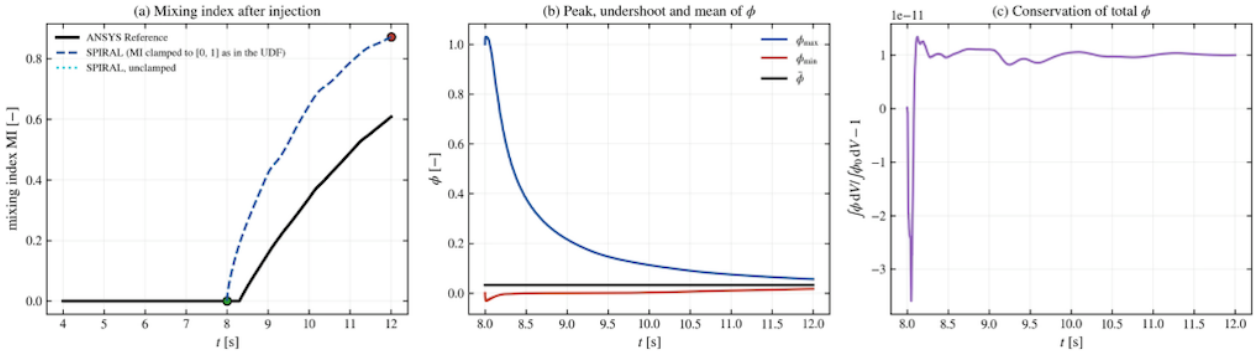






Results at same Diffusivity of ANSYS





SPIRAL in 3D Lid-Driven Cavity

Item	SPIRAL-3D as implemented in ldc3d_spiral.py
Hidden layer	One layer of P tanh nodes, tanh(w_k . x + b_k). w_k = s_k d_k with s_k log-uniform in [2, 3.5 N], d_k a random unit vector; x0_k uniform in the slab; b_k = -x0_k . w_k. Frozen, never trained. One layer shared by all outputs.
Output layer	3 linear nodes A_x, A_y, A_z (vector potential; stream function in the 2D version), no activation and no output bias. The output weights z_{c,k} are the only unknowns.
Velocity	u = discrete curl of A on the staggered grid, so the divergence is zero to round-off for any weights (measured max |div u| = 6e-27 in a 12x12x2 test run).
Physics residual	Curl of the momentum equations on the grid edges (pressure eliminated); r = 0 is the target. Time stepping BDF2.
Solver	Newton / Levenberg-Marquardt in kernel (dual) form: K = S G S^T + lambda I, Cholesky, chord re-use while the flow changes slowly; stop when the relative update is below 1e-8.
Size rule	P = ratio * n_A / 3 features per output with ratio = 2, so 3P = 2 n_A weights for n_A equations (minimum-norm solution).
Scalar	Second network at the cell centres for the passive scalar; conservative finite-volume advection-diffusion with 2nd-order linear upwind, BDF2, kernel-form solve.
Validation	Lid-driven cavity at Re = 1000 against Ghia et al. (1982) and ANSYS Fluent.
Gap against the literature	No paper in this table combines a stream function / vector potential (exact divergence-free velocity), a kernel-form Newton solve, the cavity at Re = 1000 checked against Ghia and ANSYS, and passive-scalar mixing. Only a limited search was done: confirm before claiming it in the thesis.

SPIRAL: Comparison with Recent related Random Feature / Extreme-Learning Machine PDE Solver

#	Authors	Year	Journal / venue	What they did	Similar to SPIRAL	Similarity to SPIRAL
1	Dwivedi, V.; Srinivasan, B.	2020	Neurocomputing	First physics-informed extreme learning machine (PIELM). Hidden weights are random and fixed, the PDE is imposed at collocation points, and the output weights come from one linear least-squares solve.	Frozen random hidden layer; linear output layer; solved by linear algebra, no gradient training.	High (network)
2	Dong, S.; Li, Z.	2021	Computer Methods in Applied Mechanics and Engineering	Local ELM (locELM): many small random-weight networks on subdomains joined by continuity conditions. Linear and nonlinear PDEs solved by least squares, with Newton-type iteration for nonlinear problems.	Frozen random features; linear output solve; Newton-type iteration for nonlinear PDEs.	High (solver idea)
3	Calabro, F.; Fabiani, G.; Siettos, C.	2021	Computer Methods in Applied Mechanics and Engineering	ELM collocation for elliptic PDEs with sharp gradients, compared with classical discretisations.	Frozen random hidden layer; relevant to the thin boundary layer at the lid.	Medium-high
4	Fabiani, G.; Calabro, F.; Russo, L.; Siettos, C.	2021	Journal of Scientific Computing	ELM for nonlinear PDEs plus bifurcation analysis. As read from the abstract, the nonlinear solve is a Newton-type iteration on the output weights.	Newton-type iteration on the output weights of a frozen random network.	High (nonlinear solve)
5	Dong, S.; Yang, J.	2022	Journal of Computational Physics	How to choose the range of the random hidden weights (ELM hyperparameter), with comparison against classical and high-order finite elements.	Directly relevant to SPIRAL's choice of the weight-steepness range s_min to s_max.	High (hidden-layer design)
6	Dong, S.; Yang, J.	2022	Computer Methods in Applied Mechanics and Engineering	Variable-projection method: the linear output weights are eliminated and the hidden weights are optimised. Tested on Poisson, advection, nonlinear Helmholtz and viscous Burgers (first page only was read).	One hidden layer; linear output weights solved exactly.	Medium
7	Chen, J.; Chi, X.; E, W.; Yang, Z.	2022	Journal of Machine Learning (Global Science Press)	Random feature method (RFM): random features on subdomains with a partition of unity, PDE imposed by collocation, boundary conditions by a penalty, least-squares solve.	Random-feature representation; least-squares / Newton-type solve.	Medium-high
8	Shang, Y.; Wang, F.	2024	Journal of Engineering Mechanics	Randomized neural networks with a Petrov-Galerkin (weak-form) discretisation, applied to linear elasticity and Navier-Stokes.	Random hidden layer; covers Navier-Stokes.	Medium-high (nearest on Navier-Stokes)

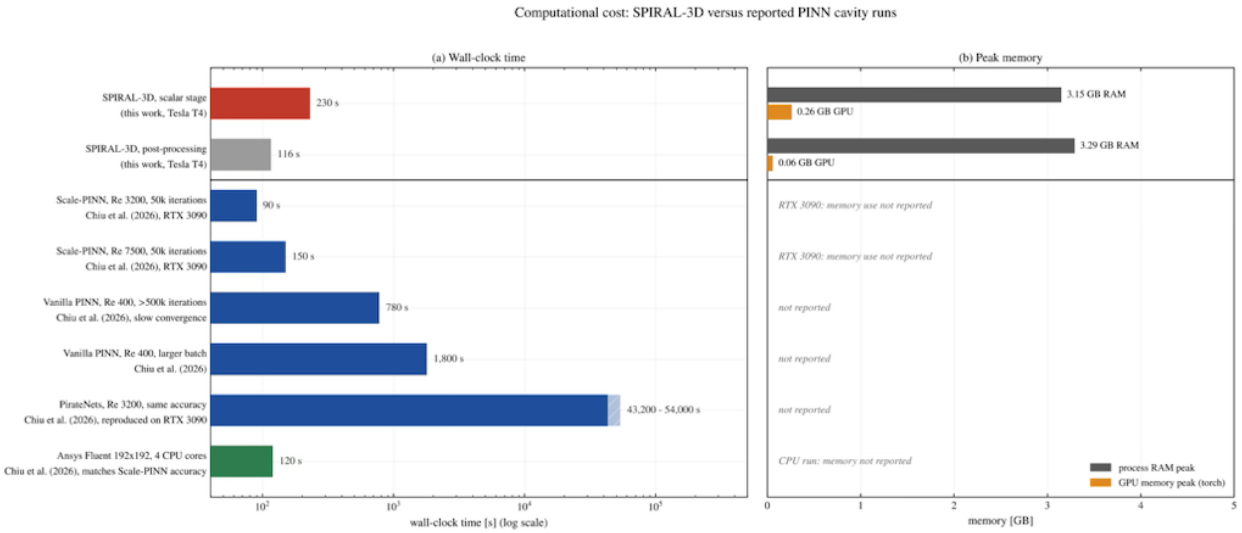
PINN vs SPIRAL

PINN studies of the lid-driven cavity compared with SPIRAL

What each paper found, and how it relates to my network

Author, year	What they found	Similarity to SPIRAL
Wang et al. (2021) *SIAM J. Sci. Comput.*	Cavity at Re = 100 only. Velocity–pressure PINNs all failed. A stream-function PINN reached 3 % relative L2 error against their own finite-difference solution, but only after re-weighting the loss terms and changing the network. Trained with 40,000 Adam iterations.	Same idea of a stream function, so continuity holds exactly. SPIRAL needs no loss weights and no gradient training, and the script targets Re = 1000.
Roy et al. (2025) *preprint, finite-difference PINN*	Re = 400 and 1000. The residual comes from central differences on a grid instead of autodiff; weights trained with L-BFGS. At Re = 1000 the lower-left vortex sits at y = 0.025 (standard) or 0.054 (sub-domain) against 0.078 in Ghia. No velocity-profile error against Ghia is given.	Closest in the residual: finite differences on a grid, as in SPIRAL. The difference is that it still trains every weight and can stall in local minima, while SPIRAL solves only the output weights with Newton.
Hu & Senocak (2025) *preprint*	Re = 2500 to 7500 with a pressure-Poisson objective and equality constraints. Needs 100,000 L-BFGS epochs and a tuned viscosity schedule. Agreement with Erturk et al. is judged by eye, with no error numbers. The authors put the cost at about two orders of magnitude above a conventional solver.	Reaches higher Re than my test case, but only qualitatively and at high cost. SPIRAL drives the residual to near zero with a few Newton iterations per step and can be checked against Ghia in numbers.
Sheikholeslami et al. (2026) *Results in Engineering, review*	Surveys PINN work on the cavity. Studies are hard to compare because architectures, boundary treatment and error metrics differ, and ablation studies are often missing.	Good reference for the state of the art. SPIRAL's grid-based residual falls in the review's "discretisation-based residual" group.
Sheikholeslami (2025) *Chalmers licentiate thesis*	A plain PINN struggles at the top corners where the lid meets the walls. Loss weights that vary in space helped most: error in u down 81 %, in v down 78 %. Forcing the lid condition with a trial function made it worse.	The lid corners are a known weak point for PINNs. In SPIRAL the lid is a ghost-cell condition in the discrete operator, so the network needs no special corner treatment.

Ghia = Ghia et al. (1982). Findings are as the authors report them. The review and the thesis were read from their abstract pages only.



All literature times are TRAINING times from Chiu et al. (2026, arXiv:2602.19475, preprint), 2D steady lid-driven cavity. Scale-PINN is that paper's own accelerated method (small batches, residual smoothing; final relative L2 error about 2-3e-2), not a typical PINN.
SPIRAL-3D: 30x30x3 cells, 3D transient with scalar; times read off my cost plot. The flow stage is missing from that plot, so the SPIRAL total is higher than the two bars shown. Different problem, mesh and hardware: indicative only.